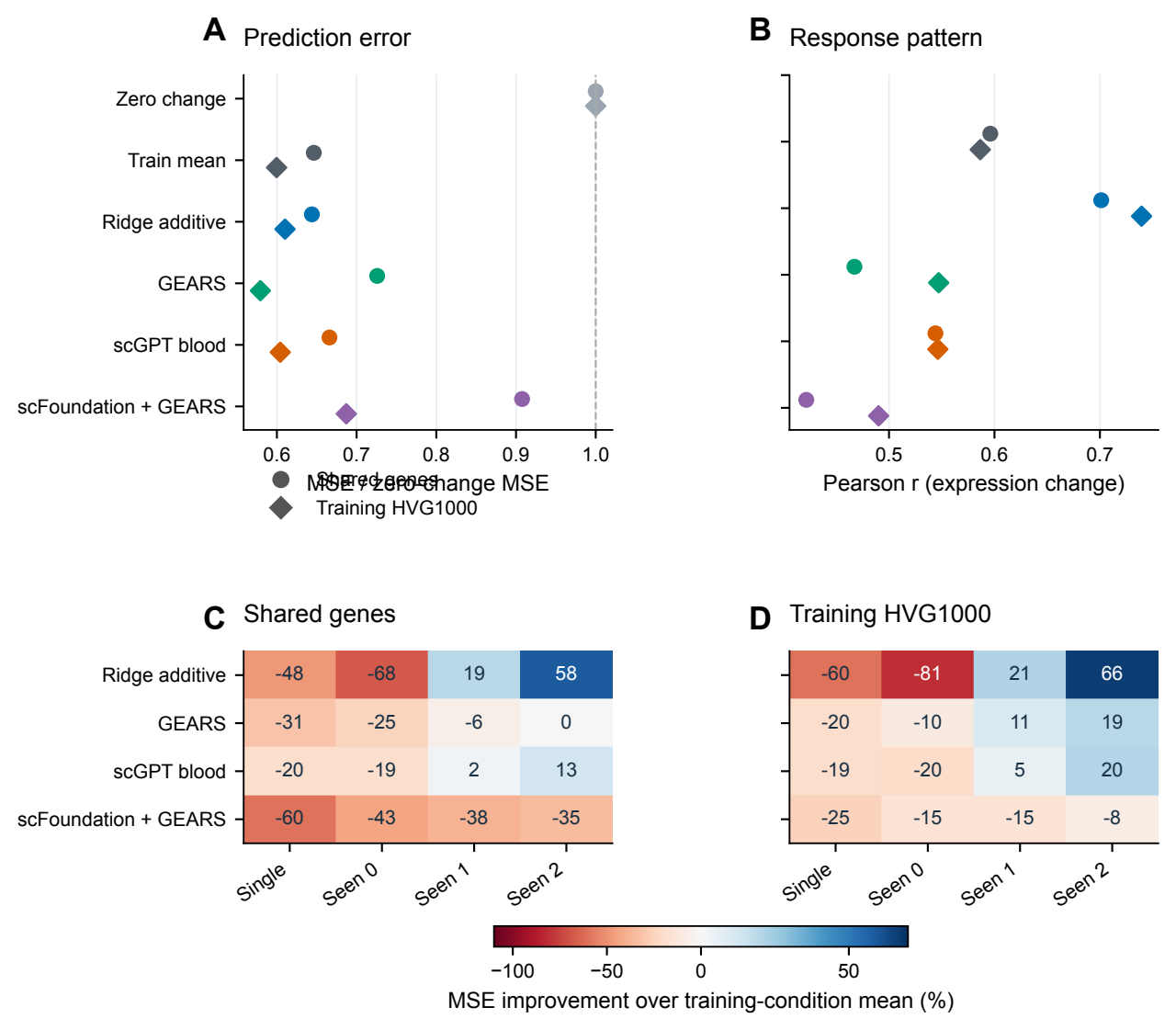
A
Prediction error
Zero change
Train mean
Ridge additive
GEARS
scGPT blood
scFoundation + GEARS
0.6
0.7
0.8
0.9
1.0
Shared genes
MSE / zero-change MSE
Training HVG1000
B
Response pattern
0.5
0.6
0.7
Pearson r (expression change)
C
Shared genes
-48
-68
19
58
-31
-25
-6
0
-20
-19
2
13
-60
-43
-38
-35
Single
Seen 0
Seen 1
Seen 2
D
Training HVG1000
-60
-81
21
66
-20
-10
11
19
-19
-20
5
20
-25
-15
-15
-8
−100
−50
0
50
MSE improvement over training-condition mean (%)